

$$x + y + z = 3$$

3 vars

$$x - y + z = 1$$

2 eqn

∞ sol's.
expectation

Guess or use least squares

$$x = 1 \quad y = 1 \quad z = 1$$

$$x = 3 - y - z$$

$$(3 - y - z) - y + z = 1$$

$$-2y = -2 \quad y = 1$$

$$x = 3 - 1 - z = 2 - z$$

General sol. $y = 1 \quad x = 2 - z$

~~GA~~

$$\begin{pmatrix} 2 - z \\ 1 \\ z \end{pmatrix}$$

vector
form

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$x + y + z = 3$$

$$x - y + z = 1$$

One Sol. $x=1 \quad y=1 \quad z=1$

$$A \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$A \cdot \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + v \right) = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

iFF $A \cdot v = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Subtract
from

$$1 + 1 + 1 = 3$$

$$1 - 1 + 1 = 1$$

Subtract

From

$$(1+x) + 1 + (1-x) = 3$$

$$(1+x) - 1 + (1-x) = 1$$

$$(x)$$

$$(x)$$

$$+ (-x) = 0$$

$$+ (-x) = 0$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}$$

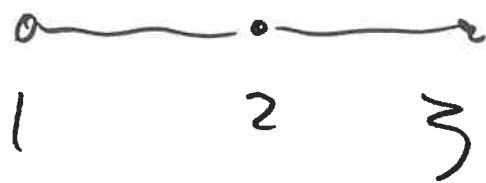
$$\text{Null Space}(A) = \{v \mid Av = 0\}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} x + y + z &= 0 \\ x - y + z &= 0 \end{aligned} \Rightarrow \begin{aligned} x &= -y - z \\ y &= 0 \end{aligned}$$

$$\text{Null}(A) = \left\{ \alpha \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Spanned by 1 vector



$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

↑

1 → 2 → 3

2-1-2

2-3-2

$a_{ij} = 1$ if edge from j to i

$a_{ij} = a_{ji}$ if symmetric
if undirected graph

Go from 3 to 5

in one step a_{53} 5th row 3rd col

in two steps $3 \rightarrow 1 \rightarrow 5$ or $3 \rightarrow 2 \rightarrow 5$ or $3 \xrightarrow{k} 5$
edge $3 \rightarrow 1$ and $1 \rightarrow 5$? = $a_{31} a_{15}$ 0 or 1

$$a_{31} a_{15} + a_{32} a_{25} + a_{33} a_{35} + a_{34} a_{45} + \dots$$

$$a_{51} a_{13} + a_{52} a_{23} + a_{53} a_{33} + a_{54} a_{43} + \dots$$

$$= \sum_{k=1}^n a_{5k} a_{k3}$$

$$\text{if } B = A^2 = A \cdot A \text{ then } b_{53} = \sum_{k=1}^n a_{5k} a_{k3}$$

Counts path of length 2

A^3 counts paths of length 3

$$a_{5k} a_{kl} a_{l3}$$



$$W + X + Y + Z = 0$$

$$W + X - Y + Z = 0$$

$$Y = 0$$

$$W + X + Z = 0$$

$$\begin{pmatrix} W \\ X \\ 0 \\ -X-W \end{pmatrix}$$

2×4
 expect rank 2
 expect Null $4-2=2$

$$= W \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} + X \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}$$

+ Guess
 or
 1st squs
Sol.

Not only way.

$$= \alpha \begin{pmatrix} 1 \\ 1 \\ 0 \\ -2 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$$

$$\alpha = W + X$$

$$\beta = W - X$$

or

$$X = \frac{\alpha - \beta}{2} = \frac{1}{2}\alpha - \frac{1}{2}\beta$$

$$W = \frac{\alpha + \beta}{2}$$

General Sol (Null 2D)

$$X_0 + \alpha V_0 + \beta V_1$$

$$\begin{bmatrix} 0.7 \dots \\ -1 \dots \\ 1 \dots \\ -0.4 \dots \\ -5 \end{bmatrix} + \alpha \begin{bmatrix} 0.5 \dots \\ 0.6 \dots \\ 0.4 \dots \\ 0.1 \dots \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -0.7 \dots \\ 0.1 \dots \\ 0.6 \dots \\ 0.1 \dots \\ 0 \end{bmatrix}$$